

LLaSA: A Sensor-Aware LLM for Natural Language Reasoning of Human Activity from IMU Data

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ABSTRACT

Wearables generate rich motion data, yet current systems only classify what happened—failing to support natural questions about why it happened or what it means. We introduce LLaSA (Large Language and Sensor Assistant), a compact 13B model that enables ask-anything, open-ended question answering grounded in raw IMU data. LLaSA supports conversational, context-aware reasoning—explaining the causes of sensor-detected behaviors and answering free-form questions in real-world scenarios. It is tuned for scientific accuracy, coherence, and response reliability. To advance this new task of sensor-based QA, we release three large-scale datasets¹: SensorCaps, OpenSQA, and Tune-OpenSQA. Together, these resources define a new benchmark for sensor-language models. LLaSA consistently produces interpretable, causal answers and outperforms commercial LLMs across both public and real-world settings.

1 INTRODUCTION

Imagine asking a conversational AI: “*Did I show signs of abnormal gait while climbing stairs? If so, why?*” A meaningful, data-driven response could offer instant insights—grounded in real-time sensor data—without requiring any technical expertise. Today, over 19 billion IoT devices, including wearables and smartphones, continuously collect motion data [35]. Yet, up to 73% of this information goes underused due to the complexity of interpreting this data [1]. This gap limits everyday decision-making in health, fitness, and rehabilitation, and also curtails richer interactions in AR/VR that increasingly rely on contextual, high-frequency sensor streams.

Advances in Large Language Models (LLMs) and Visual Question Answering (VQA) have shown how multimodal systems can improve human interaction with complex data [14, 15, 17, 24]. By fusing visual and textual information, these models excel at tasks like image captioning and scene interpretation [4, 39, 43]. In contrast, sensor-based question answering (SQA) remains underexplored, as IMU signals are fundamentally different—dense, continuous, and lacking semantic structure.

¹<https://github.com/BASHLab/LLaSA>

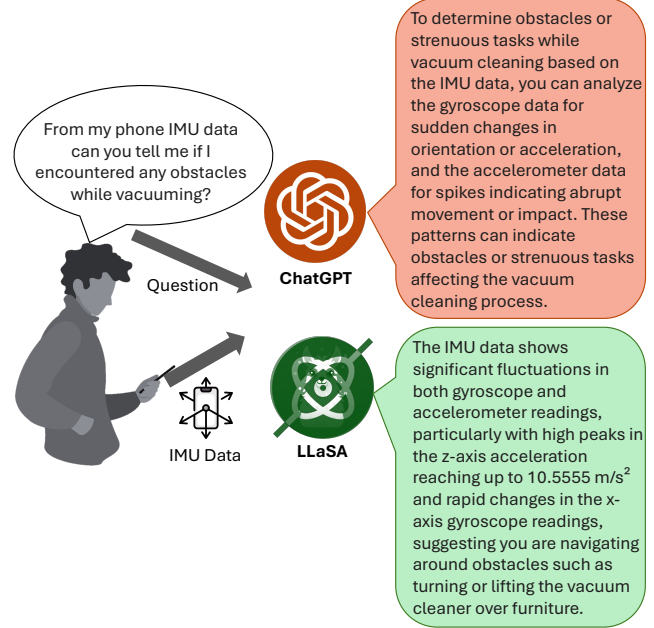


Figure 1: A real example comparing GPT-3.5 and LLaSA on an open-ended walking-pattern query. GPT-3.5 responds generically, while LLaSA grounds its explanation in IMU data—referencing upward acceleration and gyroscope stability to infer balanced movement.

Unlike images or videos, which contain visual cues with inherent semantic meaning, IMU data consists of long, continuous streams of low-level numerical values—noisy, unsegmented, and difficult to interpret. While models [5] can classify basic actions like walking or running, they often act as black boxes and provide little insight into why a particular activity occurred. Bridging this gap requires aligning raw, context-dependent time-series patterns with human-understandable concepts—especially for open-ended, causal questions. The challenge is further compounded by the fact that meaningful motion patterns often span hundreds or thousands of time steps, making it difficult to preserve temporal structure within the limited context window of language models. Addressing these limitations demands a system that can reason over long, variable-length sensor sequences, compress them into interpretable representations, and respond

with grounded, natural language explanations—without overwhelming memory or computational resources.

Effective open-ended reasoning over sensor data requires more than architectural advances—it also demands a shift in training objectives. Most existing methods, such as SensorLLM [23], are optimized for classification accuracy using discrete label prediction. While sufficient for recognition tasks, these objectives fall short when the goal is to explain why something occurred or to produce coherent, context-aware responses. Addressing this gap calls for a tuning approach that promotes causal reasoning, narrative clarity, and the ability to generate meaningful explanations grounded in sensor signals.

To bridge the gap between raw sensor data and natural language understanding, we introduce LLaSA (Large Language and Sensor Assistant), a multimodal framework that enables open-ended, context-aware question answering on IMU signals. Unlike traditional models that perform isolated activity classification, LLaSA supports natural, conversational queries and generates interpretable, causal explanations grounded in motion data. When given a user query, LLaSA processes IMU input, encodes it using a sensor-aware backbone, and aligns it with language representations to reason across modalities. As shown in Figure 1, when asked to assess a walking pattern, GPT-3.5 provides a generic response with no reference to the actual signal. In contrast, LLaSA identifies consistent upward acceleration and stable gyroscope behavior, concluding that the user ascended the stairs with control and balance. This highlights LLaSA’s ability to reason over sensor dynamics and provide specific, data-driven, and contextually relevant insights.

A core component of LLaSA is its tuning strategy, designed to improve context-aware reasoning over sensor data. Instead of optimizing for label accuracy, we train the model to prioritize three key qualities: scientific accuracy, narrative coherence, and response reliability. This process is guided by a GPT-based scoring metric that evaluates response quality and refines the model’s ability to generate grounded and interpretable explanations.

Progress in sensor-based QA has been limited by the absence of high-quality datasets that align raw IMU signals with natural language understanding. To bridge this gap, we release three large-scale resources: (1) SensorCaps, a dataset of 35,960 IMU sequences paired with descriptive captions to support motion pattern understanding; (2) OpenSQA, a collection of 179,727 question–answer pairs designed for causal and explanatory reasoning; and (3) Tune-OpenSQA, a fine-tuning set of 19,440 instruction-following examples curated to optimize accuracy, coherence, and reliability. Together, these datasets define a new benchmark for evaluating sensor-language models and support robust, interpretable reasoning in interactive AI systems.

Key contributions of this paper is summarized as follows:

- We introduce LLaSA, a the first sensor-aware open-source LLM that enables open-ended, context-aware question answering over continuous IMU sensor data, moving beyond traditional activity classification. LLaSA encodes IMU data and uses a projection-layer to efficiently maps high-frequency sensor signals into a shared representation space, allowing LLMs to process and interpret sensor-derived behaviors within a 2048-token context limit, significantly reducing computational costs.
- Unlike prior approaches that rely on classification loss, we propose a dedicated tuning strategy optimized for scientific accuracy, narrative coherence, and response reliability, using a GPT-based quality metric to refine model responses. Even without tuning with activity labels, LLaSA can generalize beyond predefined tasks, performing zero-shot activity classification and interpreting unseen activities based on motion dynamics and contextual cues.
- We introduce three large-scale datasets: (1) Sensor Captioning, pairing IMU sequences with natural language descriptions; (2) Open Sensor-aware QA, enabling reasoning beyond classification; and (3) Tuning for Open SQA, optimizing models for high-quality sensor-based explanations.

We evaluate LLaSA against GPT-3.5, Vicuna-13B, GPT-4o, and GPT-4o-mini across multiple criteria: scientific depth, narration quality, and contextual reliability. LLaSA outperforms GPT-3.5 and Vicuna-13B and surpasses GPT-4o-mini in sensor-grounded reasoning tasks. Human evaluations ($n = 22$) confirm a strong preference for LLaSA’s sensor-aware responses. In real-world testing with 10 participants, LLaSA successfully interprets unseen activities and adapts to diverse environments. It also achieves competitive zero-shot classification accuracy—without task-specific fine-tuning. Ablation studies reveal that lower LoRA ranks improve sensor–language integration, scaling enhances reasoning, and our GPT-based quality metric is a more reliable tuning signal than loss alone.

2 LLaSA ARCHITECTURE

This section describes the architecture of LLaSA as illustrated in Figure 2. LLaSA takes raw sensor data and a user query as input and generates a context-aware language response. LLaSA consists of four key components: (1) A sensor encoder that converts raw IMU sensor data into a compact feature representation, reducing context length. (2) A text encoder that transforms the input query into a query embedding. (3) A projection layer that aligns the sensor and query embeddings into a shared representation space. (4) A sensor-integrated language model that takes the combined embeddings from sensor and query to generate the final response.

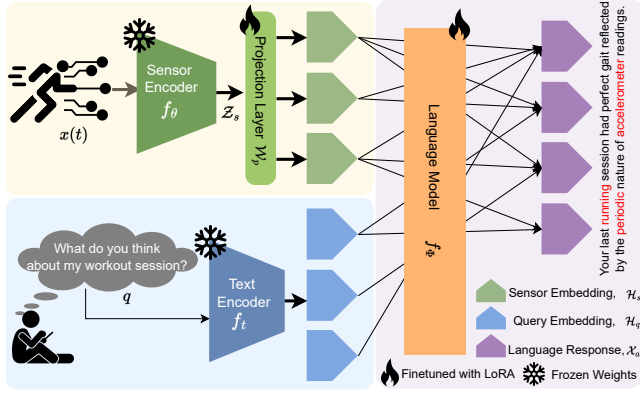


Figure 2: Architecture of LLaSA: Raw IMU signals are encoded by a frozen sensor encoder (f_θ), then projected into a shared embedding space ($\mathcal{H}_s$) aligned with the query embedding ($\mathcal{H}_q$). The language model (f_ϕ) processes both embeddings to generate the response ($\mathcal{X}_a$). For instance, given a query about a workout, LLaSA identifies the activity (running) and justifies it using periodic accelerometer patterns.

2.1 Sensor Encoder f_θ

Given raw sensor data $x(t) \in \mathbb{R}^{C \times T}$, where C is the number of channels and T is the total time period, we use a pre-trained sensor encoder with an embedding size of $\bar{k}$, denoted as $f_\theta(\cdot)$, to extract meaningful sensor representations while reducing the context length. The encoded sensor feature representation is given by: $\mathcal{Z}_s = f_\theta(x(t)) \in \mathbb{R}^{\bar{c} \times \bar{k}}$. As our sensor encoder, we employ pre-trained LIMU-BERT [41], a multi-head attention-based encoder-decoder architecture. The encoder processes raw IMU data into contextualized representations while the decoder reconstructs the original signal from these representations. The latent space representation $\mathcal{Z}_s$ captures temporal dependencies and cross-channel interactions using self-attention layers.

2.2 Text Encoder f_ϕ

A text encoder $f_\phi(\cdot)$ then converts the input query Q into a query embedding $\mathcal{H}_q \in \mathbb{R}^{\bar{c} \times k}$, where k represents the embedding dimension. As the text encoder, we use the SentencePiece [19] tokenizer included in Llama [38]. It utilizes the byte-pair encoding (BPE) algorithm [33].

2.3 Projection Layer and Alignment for Sensor-Integrated Language Models $\mathcal{W}_p$

Integrating sensor data into large language models (LLMs) requires a robust projection mechanism to align sensor embeddings with the pre-trained linguistic feature space. Given

a sensor feature $\mathcal{Z}_s \in \mathbb{R}^{\bar{c} \times \bar{k}}$, where $\bar{k}$ denotes the sensor feature dimension, we employ a learnable projection function: $\mathcal{H}_s = \sigma(\mathcal{W}_p \mathcal{Z}_s)$, where $\mathcal{W}_p \in \mathbb{R}^{\bar{k} \times \bar{k}}$ is a trainable matrix and $\sigma(\cdot)$ is the GELU non-linearity. The projected sensor representation $\mathcal{H}_s$ is concatenated with the query embedding $\mathcal{H}_q$ to form the model input: $\mathcal{H}_{\text{input}} = [\mathcal{H}_s; \mathcal{H}_q]$. This alignment ensures sensor-derived features are semantically meaningful within the LLM’s linguistic space. The projection layer, inspired by multimodal alignment techniques, functions as a sensor-adaptive tokenizer that refines sensor features into a compatible embedding space. To maintain contextual coherence, the projection is optimized to preserve semantic relationships between sensor data and text, enabling effective multimodal reasoning without compromising the LLM’s linguistic capabilities.

2.4 Sensor-Integrated Language Model

The sensor-integrated language model builds on the projected embeddings $\mathcal{H}$ to enable natural language reasoning over sensor data. Inspired by LLaVA [25], the transformed sensor embeddings replace conventional visual embeddings used in vision-language models.

Finally, the combined representation $\mathcal{H}$ is fed into the LLM f_ϕ to generate a sensor-aware language response:

$$\mathcal{X}_a = f_\phi([\mathcal{H}_s; \mathcal{H}_q]). \quad (1)$$

During autoregressive decoding, the model generates each output token conditioned on the multimodal input:

$$P(\mathcal{X}_a | \mathcal{H}_{\text{input}}) = \prod_{t=1}^T P(x_{a,t} | [\mathcal{H}_s; \mathcal{H}_q], x_{a,<t}; \phi). \quad (2)$$

where $x_{a,t}$ represents the generated token at timestep t , and ϕ denotes the LLM parameters. The objective is to maximize the probability of generating an accurate, sensor-aware response given the context.

3 LLM OPTIMIZATION FOR SENSOR CONTEXTS

Integrating sensor data into language models presents unique challenges not addressed by standard NLP tuning. Unlike text-only tasks, sensor-LLMs must handle noisy or missing signals, preserve temporal patterns, and maintain coherence in open-ended responses. Applying generic hyperparameters often results in vague or incorrect outputs disconnected from sensor context. We address this gap with a targeted tuning strategy that improves the reliability and interpretability of sensor-grounded answers—without introducing new model architectures. Our approach combines hyperparameter search with structured evaluation to promote causal reasoning and context alignment.

Traditional metrics like BLEU and ROUGE fall short in this setting [10, 12]. As shown in Table 1, they reward lexical overlap even when answers lack reasoning. In contrast, GPT-based scoring captures contextual depth and logic, providing a more meaningful way to tune and evaluate sensor-LLMs [10, 12].

3.1 Hyperparameter Optimization Strategy

Sensor-aware LLMs require hyperparameters that support both semantic fluency and signal-level reasoning. We tune five key parameters: learning rate (η) for stable adaptation across sensor contexts; batch size (B) to balance overfitting and memory use; weight decay (λ) to regularize noisy patterns; LoRA rank (r) to control adaptation capacity; and dropout (p_{drop}) to improve robustness under uncertainty. Guided by prior work [32], we constrain the search space and assess performance using our structured scoring framework (Section 3.4), ensuring alignment between sensor inputs and explanatory language output.

3.2 Limitations of Standard NLP Metrics

Lexical similarity metrics like BLEU and ROUGE fall short in evaluating sensor-aware LLMs. They rely on surface-level word overlap, missing the semantic and causal depth needed for multimodal reasoning. These metrics often misjudge semantically correct but lexically different responses and fail to account for context sensitivity—critical for interpreting temporal patterns, activity shifts, and noisy signals. Moreover, their dependence on static reference answers limits scalability in open-ended, sensor-driven tasks.

3.3 LLM Evaluation of Sensor-Aware QA

To overcome the limitations of lexical metrics, we adopt an LLM-based evaluation framework that scores responses on logical coherence, sensor grounding, and contextual relevance—rather than surface text overlap. Unlike BLEU and ROUGE, which reward token similarity, GPT-based scoring captures semantic correctness and reasoning quality.

As shown in Table 1, traditional metrics often rate shallow answers highly due to phrasing overlap, while GPT scoring favors responses with causal justification and signal alignment. This highlights the need for evaluation methods that reflect the demands of sensor-based, open-ended QA.

3.4 Scoring Framework

To evaluate sensor-aware QA beyond surface-level matching, we introduce a structured scoring framework based on LLM-generated assessments and domain-specific criteria. Each response is rated from 0 to 100 across four dimensions: correctness (alignment with expected sensor behavior), completeness (coverage of relevant signal attributes),

Table 1: BLEU and ROUGE miss reasoning. GPT-based scoring captures coherence, activity relevance, and completeness in sensor-grounded QA.

Example Query	Generic Response	Informed Response
What is the person doing?	"The person is moving." BLUE/ROUGE: High (lexical match) GPT: Low (lacks specificity)	"Smooth gait suggests walking." BLUE/ROUGE: Medium GPT: High (sensor-aware)
Is the activity classification correct?	"Yes, it matches the label." BLUE/ROUGE: High (match) GPT: Low (no reasoning)	"Patterns match walking signature." BLUE/ROUGE: Medium GPT: High (causal reasoning)

consistency (logical coherence with sensor context), and helpfulness (practical informativeness). Using GPT-4o and structured prompts² with reference answers, sensor placement, and activity labels, we compute the quality score as:

$$Q_{\text{final}} = \frac{Q_{\text{correctness}} + Q_{\text{completeness}} + Q_{\text{consistency}} + Q_{\text{helpfulness}}}{4} \quad (3)$$

By capturing reasoning and context, this framework offers a more meaningful alternative to lexical scoring.

Validation of Quality Score Reliability. To assess the reliability of our quality scoring, we asked GPT-4o and Deepseek-V3 to independently rate a subset of responses. We then computed the intraclass correlation coefficient (ICC) [18], which yielded a value of 0.92 ($p = 0.0001$, 95% CI: [0.73–0.98]). This high ICC indicates strong inter-rater reliability, confirming that the quality score is a valid measure of response quality.

3.5 Scaling Model and Data

We adopt a two-fold scaling approach to enhance sensor-grounded reasoning while maintaining computational feasibility. **Model scaling** adjusts architecture size to better capture fine-grained sensor patterns, balancing expressiveness and efficiency. **Data scaling** expands training with diverse sensor narratives, improving generalization to unseen activities and contexts. This structured scaling enhances LLaSA’s reasoning capabilities and supports efficient real-world use.

4 SENSOR-LANGUAGE DATASETS.

Sensor understanding goes beyond classification—it requires context, reasoning, and interpretability. Most existing datasets focus on activity labels, limiting models’ ability to explain or answer open-ended questions. We address this by releasing three open-source datasets that train models to interpret raw IMU signals in natural language. All datasets are available on GitHub (linked on the first page).

²All prompts are available at <https://anonymous.4open.science/r/LLaSA/>.

Table 2: Five datasets are used to generate OpenSQA and train LLaSA for open-ended QA, but not for classification fine-tuning—making classification zero-shot. SHL and real-world datasets are evaluation-only. Bolded activities are excluded from training and tuning. Total 188,429 activities from 95 users.

Dataset	Classes	Train	Test	Zero-shot
HHAR	bike, sit, stand, walk, ascend stairs, descend stairs	✓	×	✓
Motion	sit, stand, ascend stairs, descend stairs, walk, jog	✓	×	✓
Shoaib	sit, stand, jog, bike, ascend stairs, descend stairs, walk	✓	×	✓
UCI	stand, sit, lay, walk, ascend stairs, descend stairs	✓	×	✓
PAMAP2	lay, sit, stand, walk, run, cycle, Nordic walk, watch TV, computer work, etc.	✓	✓	×
SHL	stand, sit, walk, run, bike	×	✓	✓
Collected	sit, stand, pick object , jump, swirl , stretch , use stair, elevator , walk	×	✓	×

4.1 SensorCaps: Sensor-Aware Captioning

SensorCaps teaches models to describe and explain movement patterns beyond basic classification. Using public HAR datasets (Table 2), it provides **35,960** narrations that capture motion dynamics and context.

Dataset Structure and Processing. Each SensorCaps sample includes an activity label, IMU data (gyroscope and accelerometer), sensor location, and handcrafted features such as moving averages, FFTs, and statistical summaries (min, max, median, variance). These features help models summarize key motion patterns and improve interpretability. To reduce noise and focus on broader trends, IMU signals are downsampled to 10 Hz and rounded to six decimals. SensorCaps produces two outputs per instance: (1) a natural language summary of motion features like variability and frequency trends, and (2) a step-by-step, temporally grounded narration of the activity. Together, these outputs help models go beyond label prediction and generate interpretable, sensor-informed descriptions.

Feature Engineering and Annotation. IMU data is noisy and difficult to interpret directly. To support pattern extraction, we add statistical and frequency-based features: moving averages smooth short-term noise, FFTs capture cycles, and summaries quantify intensity and variability. Human experts created a subset of reference captions to guide prompt engineering, and LLM outputs were refined for clarity and accuracy before scaling.

Prompt Design for Sensor Narration. Generating high-quality narrations requires LLMs to reason over numerical sequences. To balance performance and cost, we use GPT-4o-mini with structured sensor features to boost interpretability. Each prompt follows a consistent format—captions are capped at 25 words and narrations at 500—ensuring outputs remain concise yet informative.

4.2 OpenSQA: Sensor-Aware Open QA

Built on SensorCaps, OpenSQA adds **179,727** QA pairs to train LLaSA for open-ended questions grounded in IMU data. While SensorCaps provides narration, OpenSQA enables reasoning and explanation—bridging raw sensor input and human-like interpretation.

Dataset Structure and Processing. Each OpenSQA sample includes an activity label, IMU data, sensor location, and SensorCaps outputs—providing structured context for generating interpretable QA pairs grounded in sensor reasoning.

Question Generation and Prompt Engineering. We use prompt engineering to guide reasoning over sensor features, generating diverse questions—from factual (e.g., “What activity was detected?”) to inferential (e.g., “What motion characteristics suggest a transition from walking to running?”). Each IMU instance yields five QA pairs, reviewed by experts for accuracy and coherence. Human-augmented examples improve consistency.

4.3 Tune-OpenSQA: QA Tuning for Sensors

Tune-OpenSQA includes **19,440** QA pairs for tuning rather than training. While OpenSQA teaches response generation, Tune-OpenSQA refines LLaSA’s ability to produce coherent, scientific, and context-aware explanations.

Dataset Structure and Processing. Tune-OpenSQA uses held-out PAMAP2 samples [31] not seen in OpenSQA, providing a clean benchmark for tuning. Each instance includes a movement label, sensor location, IMU data, and five QA pairs across diverse reasoning types. This setup evaluates LLaSA’s ability to explain and interpret sensor-based behavior in context.

Benchmarking for Tuning. Tune-OpenSQA serves as a benchmark for tuning sensor-aware reasoning beyond label prediction. It evaluates three areas: scientific reasoning (e.g., physics alignment), narrative clarity, and reliability in detecting noise or inconsistencies. This ensures LLaSA stays robust and interpretable in real-world settings.

Prompt Engineering for Structured Evaluation. Unlike OpenSQA, which trains for answering, Tune-OpenSQA prompts are used for evaluation. Each follows a structured

format and is paired with GPT-4o reference answers to benchmark reasoning, clarity, and scientific validity across hyperparameter settings.

4.4 Data Quality Assessment

We assess the generated data quality by considering LLMs as a judge. We use Deepseek-V3, Llama-3.1-8b, and Gemma-12b model as our LLMs to assess the quality of our dataset. We ask the LLMs to score the quality of each question and answer on a scale of 1 to 5 based on the label, narration, and input data. We use Deepseek-V3 via API and the other two models locally. As per Deepseek-V3, 87% of the data rated over 4.5, and 96% data are rated over 4.0. Similarly, Gemma-12b scores 73% data over 4.5 and 93% data over 4.0. However, Llama scores 100% of the data over 4.0 and 97% over 4.5. This consistency across all the LLMs proves that the generated data quality is high and sufficient to train large models.

5 IMPLEMENTATION DETAILS

LIMU-BERT training. We pretrain LIMU-BERT using unlabeled data from four standard HAR datasets: HHAR [36], UCI-HAR [30], MotionSense [26], and Shoaib [34], covering eight activities with 3-axis accelerometer and gyroscope signals. All signals are downsampled to 20 Hz for consistency. We also experimented with adding 18 activities from PAMAP2 [31], but found minimal impact on open-ended QA, indicating sufficient diversity in the core datasets. LIMU-BERT is trained for up to 1000 epochs using Adam with early stopping, converging in 8 hours on an RTX 3090 Ti. The best model is saved for downstream use in LLaSA.

Multimodal Projector Training. We train a GELU-activated MLP projector (W_p) using OpenSQA to align sensor features with LLM token embeddings. Training uses 16-bit precision on A100 GPUs (batch size 32) with Adam optimizer and a 0.0001 learning rate. A single epoch is sufficient for convergence. This setup ensures that the projector effectively aligns sensor embeddings with the LLM’s token space while minimizing computational overhead.

LLaSA Fine-Tuning. We apply LoRA [16] to Vicuna-7B-1.5 and 13B-1.5, alongside the pretrained projector, following the LLaVA-style instruction-tuning approach. By operating on encoded sensor embeddings instead of raw data, LLaSA fits within a 2048-token context, reducing memory use. The 7B model is trained on one A100 GPU, and the 13B model on two; raw input or longer contexts would exceed four.

We fine-tune for one epoch (batch size 16), systematically exploring LoRA ranks of 4–128 with scaling factor $\alpha = r$, as suggested by Lee et al. [20]. For LoRA, we test learning rates of $1e^{-4}$, $3e^{-5}$, and $1e^{-6}$; for the projector, we use $2e^{-5}$, $1e^{-4}$, and $1e^{-3}$. This targeted tuning ensures efficient adaptation and reliable reasoning grounded in sensor context.

6 OPEN-ENDED SENSOR-AWARE QUESTION ANSWERING EVALUATION

We evaluate LLaSA’s ability to generate interpretable, context-rich answers from raw IMU data. Rather than classifying actions, the model is prompted with open-ended questions that require causal reasoning and behavioral inference grounded in sensor context. Our evaluation spans three experiments: (1) automated quality scoring using our structured metric (Section 3.4), (2) human satisfaction ratings of generated answers, and (3) generalization to real-world data from a new environment, including activities not seen during training. Together, these results assess LLaSA’s ability to understand motion, explain it, and adapt to novel situations.

6.1 Comparison with Baseline LLMs

We compare LLaSA to baseline LLMs using structured quality scores across scientific depth, narration quality, and reliability (Table 3). This evaluation combines quantitative and qualitative insights to assess each model’s reasoning ability and contextual understanding.

Table 3: LLaSA achieves the highest science score and strong overall performance, outperforming GPT-3.5 and Vicuna-13b across all metrics. Despite its smaller context window, it rivals GPT-4o-mini in reliability and narration, demonstrating exceptional sensor-grounded reasoning.

Model	Context Length	Science	Narration	Reliability	Overall
LLaSA	2k	75.0	71.0	70.5	72.17
GPT-3.5-Turbo	16k	64.5	65.0	60.0	63.17
GPT-4o-mini	128k	72.0	77.0	78.0	75.67
Vicuna-13b	16k	63.4	67.0	65.4	65.27

LLaSA outperforms GPT-3.5-Turbo and Vicuna-13b across all evaluation dimensions, generating more accurate, contextual, and reliable answers grounded in sensor data. It also achieves the highest science score among all models, underscoring its strength in sensor-informed reasoning. While GPT-4o-mini excels in narration and reliability, it falls short in scientific depth—revealing the limitations of general-purpose LLMs for sensor-specific inference tasks.

Figure 3 highlights this distinction: LLaSA interprets nuanced motion patterns directly from sensor data, whereas baseline models often respond with generic suggestions. These results affirm the value of sensor-specific fine-tuning and structured evaluation strategies in building robust multimodal reasoning systems.

Despite its small 2K context window, LLaSA outperforms larger models with up to 64× more capacity—proving that with the right tuning, smaller models can deliver fast, accurate, and sensor-grounded reasoning on the edge.



Figure 3: Qualitive results of OpenSQA. LLaSA provides much more enriched and contextual answers (marked in red) in relation to the given IMU data. For example, for the second instance 'clip-walk', LLaSA points to the accelerometer reading in the y-axis to determine forward motion.

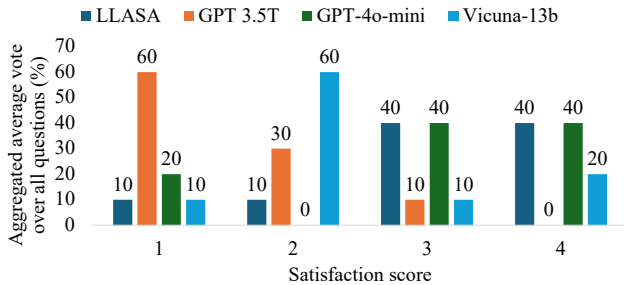


Figure 4: User satisfaction scores across 10 answers show that LLaSA outperforms GPT-3.5 and Vicuna-13b in clarity, relevance, and context. It also yields fewer low ratings than GPT-4o-mini, underscoring its strength in sensor-grounded reasoning.

6.2 User Study: Ranking QA Responses

Experimental Setup. To assess how well LLMs interpret sensor data, we conducted a survey featuring 10 QA examples across science, narration, and reliability. These examples were selected to span a range of GPT-based quality

scores, enabling a balanced evaluation. Participants ranked anonymized responses from LLaSA-13B, GPT-3.5-Turbo, GPT-4o-mini, and Vicuna-13B-1.5-16K. Each question was accompanied by a brief sensor refresher and visual plots to support contextual understanding. Unlike LLaSA, which uses encoded sensor features, GPT and Vicuna models directly analyze raw sensor signals.

Participants. We recruited 21 participants through campus boards and social media. While some had moderate familiarity with sensors, most lacked prior expertise and relied on the provided refresher to interpret the sensor plots and contextualize the QA responses.

Satisfaction Score. Participants rated each answer from 1 (least preferred) to 4 (most preferred), based on clarity, relevance, and context alignment.

Results. Human evaluations show that LLaSA outperforms GPT-3.5-Turbo and Vicuna-13b, and matches GPT-4o-mini in satisfaction, despite using a much smaller 2K context window (compared to 16K–128K). As shown in Figure 4, both LLaSA and GPT-4o-mini receive 40% of high satisfaction

scores (levels 3–4). However, GPT-4o-mini receives more low ratings, suggesting less consistent answer quality.

Participants noted that LLaSA provided clearer, more contextualized answers tied to actual sensor patterns—unlike GPT or Vicuna, which often defaulted to vague or generic analyses (see Figure 3). While users sometimes struggled to interpret raw sensor data themselves, they reported that LLaSA’s outputs improved their understanding, suggesting it successfully bridges the gap between low-level signals and intuitive explanations.

Table 4: LLaSA-7B outperforms Vicuna-13B on smartphone sensor QA, delivering more coherent and reliable responses across 9 activity classes.

Model	Model Size	Context Length	Narrative	Reliability	Overall
Vicuna	13B	16K	72.0	78.0	75.0
LLaSA	7B	2K	78.2	81.7	79.95

By grounding its answers in temporal and contextual cues, LLaSA enhances interpretability—an asset for non-expert users in domains like health monitoring, activity tracking, and human-computer interaction. These findings reinforce the importance of targeted fine-tuning and structured evaluation in sensor-aware QA models.

6.3 Generalization to Real-World Data

Data Collection. To evaluate real-world applicability, we collect a new dataset using personal smartphones from 10 users (age 18+), recruited via campus boards and social media. Participants recorded data using the Sensor Logger app [7] while performing *nine* predefined activities (Table 2) in familiar environments—70% on campus and 30% at home. The devices varied across iPhone, Samsung, and Google Pixel models, capturing a range of sensor and hardware conditions.

Each activity was performed for about two minutes in a standardized order while holding the phone. This setup introduces natural noise and variability, challenging models on the “Reliability” dimension of QA. The dataset, which reflects diverse real-world contexts, is available via our repository.

Data Processing. After filtering out the first and last 2 seconds of each reading, we split the rest into durations of 6 seconds and subsample the 100Hz readings to 20Hz, similar to the rest of the data used in our experiments. Of the approximately 20 samples per category per user, we randomly select a couple and generate 3 QA pairs of reliability and narration categories. This gives us a total of 534 QAs aware of user-collected smartphone IMU sensor data.

Evaluation. We evaluate LLaSA-7b against Vicuna-13b-1.5-16K on 50 randomly selected QA pairs per category using the quality metric from Section 3.4. These QA pairs are derived from user-collected accelerometer and gyroscope data

gathered via personal smartphones. To ensure fairness, we prompt GPT-4o to account for the variability and challenges of real-world data when scoring model outputs. As shown in Table 4, LLaSA achieves scores comparable to or exceeding those from earlier benchmarks using curated PAMAP2 data. This demonstrates LLaSA’s ability to interpret real-world sensor data and outperform larger, general-purpose LLMs—despite its smaller context window and reliance on compressed sensor features.

7 ZERO-SHOT ACTIVITY RECOGNITION

We evaluate LLaSA’s ability to recognize human activities from raw IMU data without classification-specific fine-tuning (**zero-shot**). Prompted with candidate activity labels, the model selects the most appropriate one, demonstrating its ability to generalize from open-ended sensor reasoning to structured recognition.

Prompting LLMs for HAR. We prompt the LLMs with the full set of activity labels from each dataset (Table 2). If the model fails to return a relevant label, we mark the response as “Unclear.” A label is considered relevant if it matches the dataset’s defined activity classes. To evaluate performance, we report precision, recall, and F1-score.

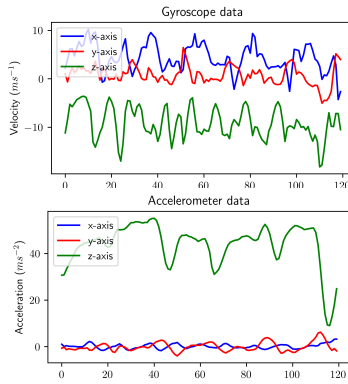
Performance Comparison and Analysis. We compare LLaSA with Vicuna-7B, GPT-3.5 Turbo, GPT-4o-mini, and a fine-tuned GPT-3.5 (GPT-3.5-T-F) trained on 5% of LIMU-BERT data. Table 5 shows both performance and context lengths across five datasets: HHAR, UCI-HAR, MotionSense, SHL, and Shoaib. SHL [13], excluded from OpenSQA training, serves as a stricter generalization test.

Table 5 shows that GPT-3.5-Turbo and GPT-4o-mini struggle to map sensor signals to activity classes in a zero-shot setting, achieving only 0.10–0.36 F1-scores when ground-truth cues are withheld. In contrast, LLaSA delivers 2.6–12× higher F1-scores across five datasets, outperforming both commercial LLMs and Vicuna-7b. Even against a fine-tuned GPT-3.5 model, LLaSA achieves 2.8× higher average F1-scores—without any classification-specific training.

Unlike GPT or Vicuna models that process raw sensor data as text, LLaSA uses compact LIMU-BERT embeddings learned during open-ended QA training—enabling it to generalize without explicit supervision. Notably, LLaSA outperforms all baselines on SHL, a dataset excluded from QA training, demonstrating strong robustness. While raw IMU data expands to 2,175 tokens, LLaSA compresses this into just 194 embeddings, keeping within a 2K context window. In contrast, baseline LLMs require 16K–128K context windows and cannot accommodate raw sensor inputs on 4×A100 GPUs, even with 8-bit quantization. This highlights LLaSA’s efficiency for resource-constrained deployments.

Table 5: Zero-shot classification performance on human activity recognition. LLaSA outperforms GPT models and Vicuna-7B with 2.6–12× higher F1-scores, despite using 8–64× smaller context windows.

LMA	Context length	HHAR			MotionSense			Shoaib			UCI			SHL		
		F-1	Precision	Recall	F-1	Precision	Recall	F-1	Precision	Recall	F-1	Precision	Recall	F-1	Precision	Recall
Vicuna-7b-16K	16k	0.05	0.12	0.03	0.09	0.15	0.12	0.08	0.11	0.11	0.04	0.25	0.03	0.08	0.38	0.07
GPT-3.5-T	16k	0.07	0.16	0.1	0.08	0.07	0.13	0.09	0.1	0.12	0.07	0.35	0.12	0.15	0.22	0.15
GPT-4o-mini	128k	0.16	0.24	0.18	0.12	0.25	0.15	0.1	0.19	0.13	0.15	0.17	0.13	0.25	0.36	0.23
GPT-4o	128k	0.07	0.14	0.08	0.08	0.26	0.06	0.14	0.35	0.15	0.04	0.16	0.03	0.22	0.27	0.23
GPT-3.5-T-F	16k	0.52	0.54	0.7	0.21	0.27	0.21	0.27	0.45	0.31	0.24	0.28	0.31	0.23	0.25	0.32
LLaSA	2k	0.84	0.88	0.86	0.83	0.85	0.83	0.81	0.84	0.81	0.72	0.82	0.75	0.65	0.76	0.71



Explain the importance of the accelerometer data in determining the upward movement of stairs and what specific changes to look for.

Worst Model

The gyroscope data shows significant fluctuations in the z-axis, with values ranging from approximately -2.5 to 2.5 radians per second. This indicates that the individual is making continuous adjustments to their body orientation to maintain balance while climbing. The rapid changes in angular velocity suggest that the person is actively compensating for shifts in their center of mass as they navigate the stairs, which is crucial for stability during the ascent.

Quality score: 30

Best Model

The accelerometer data is crucial in determining the upward movement of stairs as it captures the vertical acceleration experienced by the individual. The readings show significant spikes, particularly in the z-axis, indicating moments of upward force during the ascent. To identify these moments, one should look for peaks in the z-axis readings that exceed the average gravitational force (approximately 9.81 m/s²), which would correspond to the individual exerting upward force against gravity. Additionally, analyzing the timing and magnitude of these peaks can provide insights into the individual's rhythm and effort during the climb.

Quality score: 90

Figure 5: The best LLaSA-7B model (from Table 6) delivers a more accurate and contextually aligned response than the lowest-scoring variant. The weaker model incorrectly references gyroscope data despite the question targeting accelerometer readings—an error reflected in its GPT-assisted score.**Table 6: Hyperparameter tuning and model scaling help find the best-performing model through scientific, narrative, and reliability assessment QA Tasks.**

Model size	#QA	LR	Projector LR	LoRA Rank	Science QA Quality	Narration QA Quality	Reliability QA Quality	Overall Score
7b	131,407	1e-4		8	41.0	52.5	45.0	46.1
				16	32.5	54.0	44.5	43.7
			128	8	32.0	49.0	44.0	42.8
				2e-5	16	36.0	55.0	44.3
			3e-5	8	56.0	56.5	51.5	54.7
				16	49.5	53.5	52.0	51.7
				128	44.0	49.0	44.5	45.8
				2e-5	8	50.0	54.0	52.5
	179,727	1e-4		16	27.5	40.5	58.5	42.2
				128	32.0	38.0	53.5	41.2
			1e-3	4	36.5	45.5	34.5	38.8
				8	32.5	39.5	32.5	34.8
		3e-5	1e-6	4	35.0	45.0	47.0	42.3
				8	30.0	43.0	32.5	35.2
			1e-3	4	28.0	40.5	50.0	39.5
				8	31.0	38.5	48.5	38.2
13b	179,727	3e-5	1e-4	4	34.0	44.0	49.0	42.3
				8	75.0	71.0	70.5	72.17

8 ABLATION STUDY: HYPERPARAMETER TUNING STRATEGY

We examine how LoRA rank, model size, and dataset composition shape LLaSA’s performance across reasoning quality, multimodal alignment, and overall efficiency. A qualitative comparison highlights differences between top- and bottom-performing models in sensor comprehension. Lastly, we show that training loss fails to reflect real-world performance, while structured quality scoring proves more reliable.

8.1 Hyperparameter Tuning Experiments

We conduct a systematic study to optimize LLaSA’s performance on open-ended IMU-based QA, evaluating configurations using 30 QA pairs across three reasoning types. Table 6 outlines the explored hyperparameters: LoRA rank, model learning rate, and projector learning rate.

Our results show that lower LoRA ranks—particularly rank 8—yield better response quality across scientific reasoning, narration, and reliability. This supports findings from prior work on instruction tuning [11, 44], suggesting compact adaptation helps align sensor-language representations and reduces overfitting to noisy IMU data.

We also compare LLaSA-7B and 13B—based on Vicuna-7B and 13B, respectively—and find that the larger model consistently outperforms the smaller one in nuanced reasoning. While low-rank LoRA improves alignment efficiency, model scaling enhances interpretability and output quality.

Finally, we test the impact of increasing the training set with QAs from PAMAP2, bringing the total to 179,727. The additional data has limited effect, suggesting diminishing returns beyond a certain dataset size.

8.2 Qualitative Insights from Tuning

While quantitative metrics offer an overview of model performance, qualitative analysis uncovers key gaps in reasoning, coherence, and sensor interpretation. Figure 5 contrasts the best and worst LLaSA-7B configurations from hyperparameter tuning. The top-performing model generates precise, context-rich answers. It distinguishes between walking on flat ground and stair ascent by analyzing z-axis acceleration, correctly referencing vertical force changes and gravitational cues. This reflects strong biomechanical reasoning grounded in sensor data. Conversely, the lowest-scoring model misreads sensor patterns, attributing vertical fluctuations to lateral motion and offering vague, generic explanations. It fails to connect terrain changes to accelerometer or gyroscope signals, revealing a lack of sensor-awareness and weaker contextual grounding.

8.3 Quality Scoring Provides a More Reliable Metric Than Training Loss

Training loss is often used to guide model selection, but our findings reveal its limitations in sensor-based reasoning. Surprisingly, the worst-performing model in our study had a lower loss (1.474) than the best model (1.526), illustrating that lower loss does not imply better real-world reasoning. Token prediction accuracy alone fails to capture a model’s ability to interpret sensor data and generate meaningful responses.

In contrast, our structured quality scoring framework—based on scientific reasoning, narration, and reliability—offers a more task-relevant evaluation. It prioritizes models that exhibit contextual understanding, logical consistency, and sensor-aware insights. Unlike training loss, quality scores directly reflect the model’s interpretability and practical utility in sensor-based QA tasks, making them a more reliable metric for hyperparameter tuning.

9 RELATED WORK

LLMs have demonstrated remarkable performance across various NLP tasks, with architectures like GPT-4 [2], LLaMA [38], and PaLM [8] achieving state-of-the-art results. However, they inherently lack the ability to process non-textual

modalities, limiting their use in real-world multimodal scenarios. Recent advances have extended LLMs to include vision, audio, and sensor inputs, enabling richer contextual understanding. Models such as Flamingo [3], BLIP-2 [22], and LLaVA [25] have explored vision-language modeling by aligning images and textual embeddings. Still, they largely focus on visual data and lack robust sensor integration. While these systems support both open- and close-ended tasks, their capacity for contextualized QA does not extend to moment-level sensor reasoning. Recent work like LLMsense [29], MentalLLM [42], IMUGPT2.0 [21], and Sensor2Text [6] demonstrates long-term trace analysis or high-level event detection, but they do not offer interactive QA grounded in fine-grained, real-time sensor data.

Wearable inertial measurement unit (IMU) sensors have played a critical role in human activity recognition (HAR), particularly through self-supervised frameworks like LIMU-BERT [41], which leverages unlabeled IMU data to achieve substantial gains in recognition accuracy. Despite these advances, instruction-following remains outside the scope of traditional IMU-based HAR approaches. Prior research, such as Penetrative AI [40], applies GPT models for sensor data analysis, but its focus is primarily on detection (e.g., motion or heartbeat analysis) without extending to a QA format.

In terms of temporal data narration, approaches like WavCaps [27] have pioneered the use of GPT for generating audio event descriptions, and Gong et al. further applied such methods to construct OpenAQA, a QA dataset for audio. However, methods for translating IMU data into comprehensive narrations, particularly to support QA, remain largely unexplored. Our work extends narration frameworks to IMU data, creating SensorCaps—a wearable sensor dataset supporting open-ended QA for human activity understanding.

Furthermore, recent advances in wearable sensor data interpretation leverage LLMs for health and activity recognition tasks. Cosentino et al. [9] propose the Personal Health Large Language Model (PH-LLM), a fine-tuned version of Gemini [37], designed to process both textual and numerical health data. They demonstrate the feasibility of using LLMs to generate personalized insights and medical recommendations based on wearable data. Similarly, Merrill et al. [28] introduce the Personal Health Insights Agent (PHIA), leveraging LLM-driven retrieval and reasoning, we analyze behavioral health data, showing LLMs can effectively handle numerical and open-ended health queries.

For human activity recognition (HAR), Li et al. [23] introduce SensorLLM, a framework aligning LLMs with motion sensor data using special token embeddings and automatic trend-descriptive text generation. They show that SensorLLM, through task-aware tuning, can match or surpass conventional deep learning models in HAR tasks while also exhibiting robust generalization across diverse sensor datasets.

This work aligns closely with our research, as we also explore multimodal fusion strategies to enhance the reasoning capabilities of LLMs over wearable sensor data.

Our research builds upon these foundational efforts by extending LLMs’ ability to contextualize and interpret wearable sensor data for open-ended question-answering tasks. Unlike previous works focused primarily on classification, we developed a structured evaluation framework to assess reasoning quality, logical consistency, and contextual relevance in sensor-informed responses. Our approach emphasizes sensor-language alignment, leveraging multimodal embeddings and fine-tuning strategies to improve LLM adaptability in non-traditional modalities.

10 DISCUSSION

The development of sensor-aware multimodal LLMs opens new avenues for future research. While our work connects IMU data to human activity understanding, several challenges and opportunities remain. We outline key directions to advance LLM research for wearable sensing applications.

Multimodal Integration for Complex Environments. Enhancing activity recognition in complex environments requires integrating sensor data with additional modalities such as video or environmental inputs. Video provides contextual information that complements motion signals, enabling more accurate analysis in scenarios where IMU data alone is insufficient. This integration is vital for advancing applications in domains like smart homes and healthcare.

Scalability and Generalization to New Activities. Although effective on diverse activity sets, sensor-aware LLMs must improve in scalability and adaptability to novel, unseen activities. Future work will emphasize zero-shot and few-shot learning to enhance generalization with minimal supervision. Our evaluations on unseen datasets indicate model robustness, but expanding generalization remains a key priority.

Model Scaling and Extended Context Windows. Increasing model size and context length can significantly enhance performance. Employing parameter-efficient fine-tuning (PEFT) on models up to 70B parameters and extending context windows to 16K tokens would improve the model’s ability to capture long-range dependencies and interpret complex temporal data, particularly in motion-rich scenarios.

Conversational Capability of LLaSA. While LLaSA effectively processes contextual sensor data, it lacks conversational interaction. Introducing dialogue capabilities would enable users to query, receive explanations, and make informed decisions in real time. Future iterations will aim to transform LLaSA into a conversational, multimodal assistant to enhance user engagement and system usability.

11 CONCLUSION

We present LLaSA, a sensor-aware multimodal large language model that enables natural language reasoning over IMU data. By aligning sensor-derived embeddings with an LLM’s token space, LLaSA interprets human activity with context and precision. We introduce three supporting datasets (SensorCaps, OpenSQA, and Tune-OpenSQA) to train and evaluate the model’s ability to generate structured, insightful responses. LLaSA outperforms GPT-3.5-Turbo and Vicuna-13b, and competes closely with GPT-4o-mini, despite operating with a smaller context window and relying on sensor encoders rather than raw input. These results highlight the benefits of multimodal alignment and structured hyperparameter tuning for efficient, sensor-grounded reasoning. Our findings also show that training loss is a poor proxy for real-world performance, reinforcing the need for quality-based evaluation. Future work will extend LLaSA to support richer modalities (e.g., audio, video, environmental sensors) and explore scaling strategies to enhance long-range reasoning. These efforts will further LLaSA’s utility in domains such as healthcare, HCI, and wearable systems.

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